

Paper Practice 1.3**Properties of Real Numbers**

Show all of your work in the space provided. When a problem asks for a property, write its FULL name. Circle your final answers.

- 1.** Name the property of equality shown by each statement. (Properties of Equality)

a. If $14 + 6 = 20$, then $20 = 14 + 6$.

b. $n + 9 = n + 9$

c. If $x = 7$ and $7 = 3 + 4$, then $x = 3 + 4$.

- 2.** Name the property shown by each statement.
(Number Properties)

a. $23 \cdot 1 = 23$

b. $8 + (-8) = 0$

c. $(2/5)(5/2) = 1$

d. $17 \cdot 0 = 0$

3. Evaluate: $6 - 2^2 + 9(1)$ (Evaluate)

4. Evaluate: $(3/4)(4/3) \cdot 12$ (Evaluate)

5. Evaluate: $(18 - 2 \cdot 3 + 5) \cdot 0$ (Evaluate)

- 6.** Use the commutative and associative properties to evaluate $25 + 38 + 75 + 62$.
Show the regrouping. (Mental Math)

- 7.** Use the commutative and associative properties to evaluate $4 \cdot 17 \cdot 25$. Show the regrouping. (Mental Math)

- 8.** Use the properties to evaluate $36 + 59 + 64 + 41$. Show the regrouping. (Mental Math)

- 9.** Jade spends \$41, \$36, \$39, and \$44 on groceries over four weeks. Use one of the properties to find her total, and name the property you used. (Apply)

- 10.** Name the property used in each step. (Justify)

$$12 + 45 + 8 + 55$$

$$= 12 + 8 + 45 + 55$$

$$= (12 + 8) + (45 + 55)$$

$$= 20 + 100 = 120$$

- 11.** Error Analysis – Gus said that $(5 + 3) + 7 = 5 + (3 + 7)$ shows the Commutative Property. Identify his mistake. (Error Analysis)

- 12.** Write your own example of the Multiplicative Inverse Property, then explain in one sentence why the product is always 1. (Explain)